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Optimizing Hydraulic Fracturing Performance in Depleted Sandstone Reservoirs: The Role of Pre-Pad Solution in Fluid Recovery and Productivity Improvement

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Abstract

In the BED16C6 field, located in the Western desert, Egypt, the Abu Roash (G) sandstone reservoir has been one of the most developed reservoirs that mainly produce via hydraulic fracturing. A significant challenge associated with hydraulic fracturing in conventional depleted reservoirs is the potential formation damage induced by frac-fluid, which can impair a well's post-fracturing performance.

This paper examines the impact of injecting a pre-pad solution comprising a high concentration of oxidizing breaker, a microemulsion surfactant, and acetic acid prior to pumping the crosslinked fluid into the formation. This approach is particularly relevant when higher-than-standard gel loadings are used. The use of highly concentrated crosslinked gel fluids often results in increased gel residue, leading to greater formation damage and reduced regained permeability.

Frac-fluid formulations typically incorporate surfactants and breakers with the desired concentrations to facilitate the post frac clean-up process, reduce surface and interfacial tension, promote fluid recovery, and prevent emulsion formation. Although the concentrations of surfactants and breakers are carefully designed to effectively break the frac-fluid and help prevent emulsion formation, they may be insufficient to completely break the fracturing fluid in depleted sandstone formations, particularly under high gel loading conditions. With the concept of a pre-pad solution, high breaker concentration, high acetic acid concentration, and high surfactant concentration aims to break any remaining gel residues, therefore enhancing frac-fluid recovery and ensuring no generated emulsions, thus, leading to higher post frac productivity.

Post-frac well testing demonstrated enhanced fluid recovery and productivity in wells treated with the pre-pad technique, compared to offset wells with similar reservoir and formation characteristics. The pre-pad approach has been successfully implemented in multiple wells in the Western Desert of Egypt. This paper focuses on a specific case study in the Abu Roash (G) formation, comparing post-frac well performance with offset wells to assess the impact of the pre-pad solution. This study highlights the significant potential of pre-pad solution for optimizing hydraulic fracturing operations in challenging, depleted sandstone reservoirs.

Introduction

Hydraulic fracturing remains one of the most effective techniques to unlock hydrocarbons in low-permeability and depleted reservoirs. In the Western Desert of Egypt, the Abu Roash (G) member of the BED16C6 field has consistently been targeted for stimulation due to its tight, clastic nature and declining reservoir pressure. Historically, hydraulic fracturing treatments in this region have relied heavily on high-viscosity, crosslinked gel systems to achieve wide fracture geometry and adequate proppant placement. However, these treatments often leave behind gel residues that significantly impair formation permeability and reduce post-frac well performance.

In depleted sandstone reservoirs, several interrelated factors contribute to poor post-fracturing results: low differential pressures, reduced cleanup efficiency, increased capillary trapping, and potential emulsion formation. These issues are especially pronounced when high-polymer loading fluids are used. While conventional clean-up systems with in-situ breakers and surfactants aim to address this, their performance in such environments often falls short.

The Abu Roash (G) reservoir has shown strong dependency on clean, conductive fractures due to its relatively low matrix permeability and lack of natural fractures. Any impairment, such as residual gel or emulsion blockage, can drastically reduce the effective fracture conductivity. Field experience has shown that even small volumes of leftover polymer can plug pore throats and delay or suppress production response.

This paper investigates a novel pre-treatment technique involving a high-concentration pre-pad solution injected prior to crosslinked fluid stages. The pre-pad solution, rich in oxidizing breaker, surfactant, and acetic acid, is designed to enhance polymer degradation, mitigate formation damage, and facilitate fluid recovery, especially in scenarios where gel loadings exceed standard industry practice.

The challenge of incomplete gel clean-up in depleted sandstones has been well documented. Surfactants and breakers are traditionally incorporated into frac-fluid systems to aid in residue removal (Fan et al., 2018). However, in formations like Abu Roash (G), where pore pressures are low and capillary forces are high, standard concentrations may fail to achieve complete degradation, resulting in persistent emulsion blocks and permeability loss (Gbodi and AlTayyar, 2023).

A strategic injection of a chemically aggressive pre-pad can precondition the near-wellbore environment, ensuring that subsequent crosslinked gels are more effectively broken and removed. This methodology is particularly suited for conventional reservoirs where long, clean fractures are more desirable than complex networks (Palanivel et al., 2022). Previous implementations of similar approaches in sandstone reservoirs have highlighted improved proppant transport, enhanced fracture conductivity, and faster fluid recovery (Al Shueili et al., 2023).

This study also considers the operational flexibility of the pre-pad method. Unlike redesigning the main frac stages, implementing a pre-pad requires minimal surface equipment changes and can be easily incorporated into standard pumping schedules. Furthermore, its chemical nature is compatible with most crosslinked gel systems, making it a low-risk, high-impact solution.

In this study, we present a detailed case analysis of the pre-pad solution applied in the Abu Roash (G) reservoir. We compare performance metrics from wells treated with this technique to offset wells subjected to traditional fracturing designs. Key parameters such as fluid recovery rate, post-frac productivity index, and evidence of emulsion blockages are examined. Additionally, the paper contextualizes the findings within broader hydraulic fracturing literature to assess the general applicability of the pre-pad approach in similar geological settings.

The goal of this work is twofold: to demonstrate the operational viability and production benefits of the pre-pad breaker solution and to provide a roadmap for integrating such treatments into fracturing programs targeting depleted sandstone reservoirs globally

Geological and Reservoir Context

The Abu Roash (G) formation is part of the larger Abu Roash Group, located within the prolific Badr El Din (BED) concession of Egypt's Western Desert. This region has long been recognized for its complex stratigraphy and multilayered hydrocarbon potential. The Abu Roash (G) member, specifically, is characterized by fine- to medium-grained clastic sediments, interbedded with thin shales and occasional carbonate streaks. Deposited in a shallow marine environment during the Cenomanian age, the ARG formation generally presents moderate porosity but very low permeability, often necessitating stimulation.

Structurally, the BED16C6 field lies on a gentle anticline bounded by faulting, which provides vertical and lateral compartmentalization. The target zone typically ranges between 3,500 to 3,700 meters true vertical depth, with formation temperatures around 240–255°F and initial reservoir pressures historically ranging from 5,500 to 6,500 psi. However, many wells in this region now encounter significantly depleted pressures, with some falling below 4,000 psi.

Core data and petrophysical logs reveal average porosity in the range of 8% to 12%, with effective water saturation ranging from 20% to 30%. Matrix permeability is generally less than 2 mD, placing this formation firmly within the tight reservoir category. As a result, natural production is minimal without stimulation.

The ARG formation lacks significant natural fracturing, which makes the geometry and cleanliness of hydraulically induced fractures the primary determinant of well productivity. A clean, planar fracture with effective proppant transport and minimal residual damage is essential. This contrasts with unconventional shale plays where fracture complexity and surface area are more critical.

Operationally, the region poses several challenges. The high bottomhole static temperatures necessitate thermally stable fluids and breakers. The clay content in shales and siltstones introduces potential for clay swelling and fines migration. Furthermore, the low reservoir pressure complicates the flowback process, often requiring artificial lift or extended cleanup periods. In prior stimulation campaigns across the ARG formation, conventional crosslinked guar systems with standard oxidative breakers have led to persistent formation damage, as evidenced by poor regained permeability and delayed production response. While pulsed fracturing techniques and digital control systems have improved proppant placement ([Palanivel et al., 2022](#)), the clean-up efficiency of high-viscosity systems remains an issue. Consequently, the ARG formation presents an ideal candidate for evaluating pre-pad chemical treatments aimed at mitigating residual gel and emulsion damage.

This geological and reservoir overview provides the foundation for understanding the rationale behind applying a high-strength pre-pad breaker solution. The unique combination of tight rock properties, absence of natural fractures, and declining reservoir pressure necessitates innovative solutions to improve stimulation outcomes and long-term productivity.

Challenges with Conventional Fracturing in Depleted Reservoirs

Hydraulic fracturing in tight sandstone reservoirs is inherently complex, but the challenges multiply under depleted pressure conditions. In formations like Abu Roash (G), where natural drive mechanisms have been exhausted and the pressure differential is low, conventional fracturing systems often fall short in achieving effective cleanup and sustained productivity. The primary issues revolve around gel retention, emulsion formation, capillary effects, and ineffective fluid recovery.

One of the central problems in depleted reservoirs is polymer residue left by high-viscosity crosslinked gel systems. These gels, typically composed of guar or derivatized guar polymers crosslinked with borate or zirconium agents, are essential for proppant transport. However, in the absence of strong reservoir pressure to mobilize fluids post-fracture, these systems often fail to degrade completely ([Fan et al., 2018](#)). Even with built-in breaker packages, the amount of residual polymer trapped in the fracture face and matrix can severely reduce regained permeability.

Another critical factor is low differential pressure during flowback. In high-pressure environments, the pressure gradient drives both fluid and broken gel from the fracture and near-wellbore region. In depleted

reservoirs, this driving force is significantly reduced, resulting in longer cleanup times and higher residual water saturation. This situation is aggravated when capillary end effects dominate, trapping fluids in tight pore systems and further inhibiting flow (Gbodi and AlTayyar, 2023).

Emulsion formation is another well-documented challenge. When water-based frac fluids come into contact with formation hydrocarbons under turbulent mixing conditions, stable emulsions can form. These emulsions can plug pore throats and fracture channels, reducing effective conductivity. Conventional surfactant systems, though incorporated into most fracturing fluids, are often not optimized for emulsion destabilization under reservoir-specific conditions. In depleted formations like Abu Roash (G), where emulsion stability is exacerbated by long contact time and low turbulence, conventional surfactant doses may prove inadequate (Palanivel et al., 2022).

Breaker efficiency also plays a significant role in cleanup performance. Many conventional systems rely on oxidizing agents like ammonium persulfate or encapsulated enzymes to degrade gel structures. While these work well in lab conditions, field implementation often faces thermal degradation, premature activation, or delayed reaction due to low fluid residence time. In reservoirs with temperatures in the 235–260°F range, as in Abu Roash (G), maintaining optimal breaker activity becomes even more difficult. Fan et al. (2018) noted that in such scenarios, gel residue often remains even weeks after stimulation, compromising fracture conductivity.

In addition to these subsurface phenomena, operational factors such as screenouts, excessive pad volumes, and proppant placement inefficiencies contribute to inconsistent results. In cases where viscosity is too high, fracture width increases at the expense of length, and proppant fails to reach the desired depth. The resulting fracture geometry may be conductive near the wellbore but ineffective further out, limiting reservoir drainage.

Previous field results from Abu Roash (G) and other Western Desert fields have shown wide variation in post-frac performance, even among wells with nearly identical reservoir parameters. The common denominator in underperforming wells was poor fluid recovery, often tied to incomplete gel breaking and high-water production a trend mirrored in similar formations worldwide (Al Shueili et al., 2023).

In this context, the use of a chemically aggressive pre-pad breaker system—formulated with surfactants, oxidizers, and organic acids—offers a promising solution. By preparing the formation before gel injection, such systems can address multiple challenges simultaneously: they reduce emulsion stability, weaken gel structure at the fracture face, and improve fluid compatibility. This approach, explored in the following sections, is particularly suited to the technical and economic constraints of depleted sandstone reservoirs like Abu Roash (G).

Methodology

This section outlines the technical framework behind the implementation of the pre-pad breaker system in the Abu Roash (G) reservoir. The methodology encompasses treatment design, fluid composition, job execution, and post-frac diagnostics. The approach aimed to achieve maximum near-wellbore conditioning while preserving operational simplicity and compatibility with existing surface equipment and gel systems.

Treatment Objectives and Design Philosophy

The primary objective was to enhance post-fracturing fluid recovery and regain permeability by mitigating the common causes of damage: gel residue retention, emulsion formation, and capillary-induced water block. The pre-pad concept was introduced as a precursor to traditional crosslinked gel injection, focusing on:

- Introducing a high-concentration oxidizing breaker to pre-weaken any polymeric material.
- Along with the acetic acid added it will help improve fluid breaking and residue degradation.

- Applying a microemulsion surfactant to destabilize any oil-water emulsions and reduce surface/interfacial tension.

The sequence was carefully timed to ensure these fluids contacted the formation well in advance of the main crosslinked gel.

Pre-Pad Fluid Composition

The pre-pad fluid consisted of three key chemical agents in a water base (including liquid clay control):

1. **Oxidizing Breaker:** Ammonium persulfate at concentrations between 8.0–10.0 gal per thousand gallons (gpt), depending on expected fracture volume and fluid exposure time.
2. **Organic Acid:** Acetic acid at 5 gpt, chosen for its lower corrosion risk and compatibility with most crosslinkers compared to hydrochloric acid.
3. **Surfactant:** A proprietary nonionic microemulsion blend, effective at reservoir temperatures up to 275°F, at concentrations of 2–4 gallons per thousand gallons (gptg). This surfactant was selected based on lab tests confirming low emulsion stability and interfacial tension reduction below 10 dynes/cm.

Main Frac Stage: Crosslinked guar gel with 35–45 lb/Mgal loading, crosslinked with borate chemistry, and loaded with 20/40 mesh white proppant at concentrations ramping from 1 to 7 ppg. Pump rates ranged from 25 to 35 bpm, depending on surface constraints and fracture pressure response.

Breaker systems in the main gel included encapsulated persulfate and internal enzyme packages, but were intentionally kept at moderate concentrations, relying on the pre-pad for the majority of residue degradation.

Field Application: Case Study in Abu Roash (G)

To validate the effectiveness of the pre-pad breaker system under real field conditions, a pilot treatment was executed in a vertical well within the BED16C6 field targeting the Abu Roash (G) sandstone. The results were then benchmarked against a nearby offset well with comparable geological, operational, and historical parameters but completed using conventional fracturing fluid system without a pre-pad. [Figure 1](#) shows the 2 wells observed here in this case history where Well A was treated with prepad ahead of treatment and well B was not.

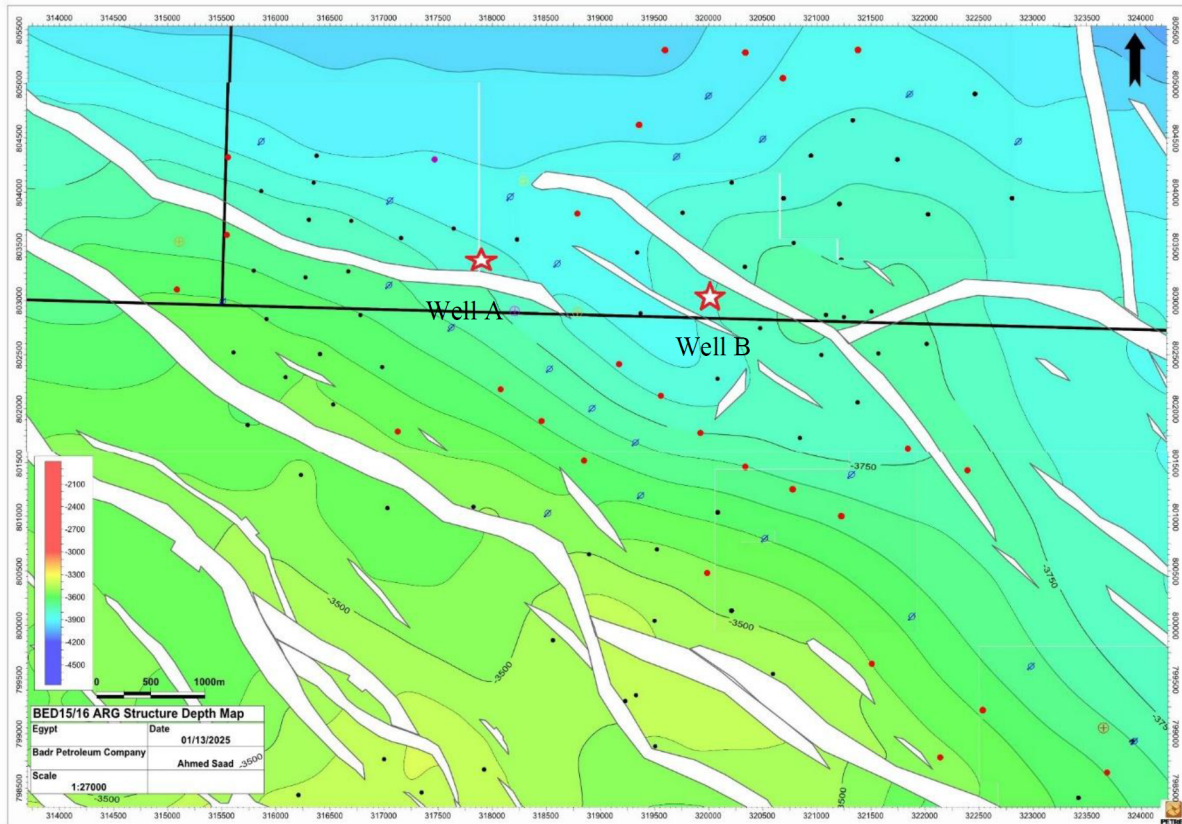


Figure 1—BED 16C6 Structure Map

Candidate Well Selection

The pilot well (Well A) and offset (Well B) were selected based on matching formation depth, pay thickness, log response, and reservoir pressure. Both wells were vertical with perforations across a net pay of approximately 7.0 meters. Petrophysical analysis showed similar porosity (~12%), permeability (0.05–1.5 mD), and minimal shale lamination in both zones.

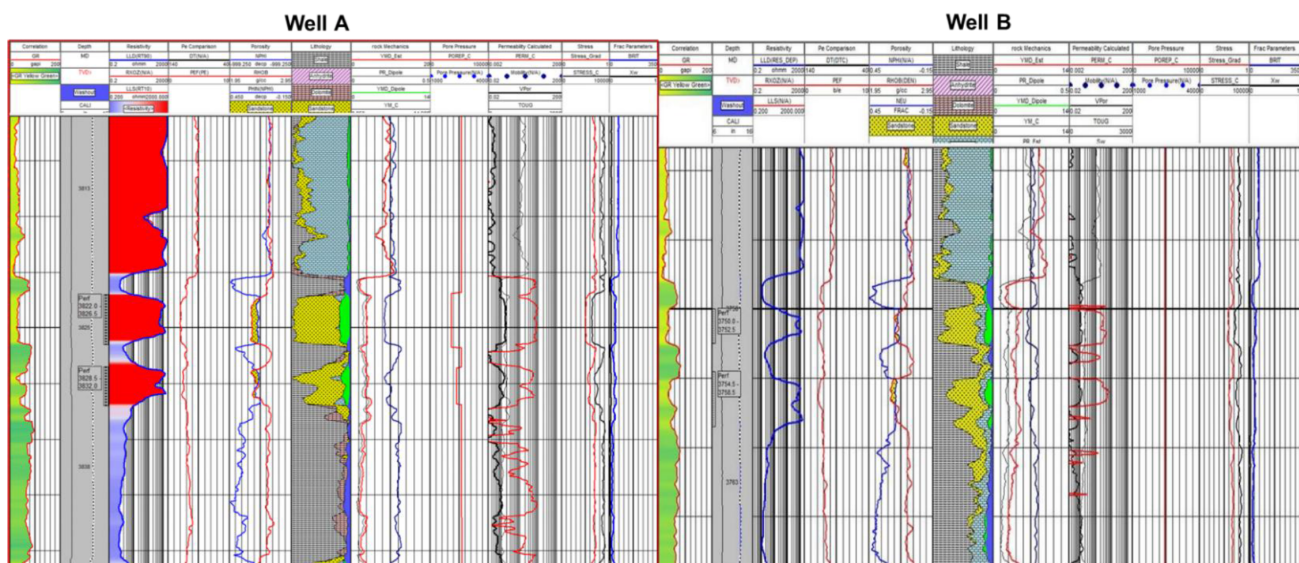


Figure 2—Petrophysical Evaluation for Well A & Well B

Reservoir pressure at the time of treatment was +/- 4,700 psi in Well A and +/-4,800 psi in Well B. Bottomhole temperature was estimated at 250°F for both wells. These parameters allowed a fair comparison between the pre-pad and non-pre-pad stimulation results.

Frac Treatment Design – Well A

Well A was treated with the enhanced pre-pad system. Key treatment parameters included as shown in figure 3 :

- Pre-Pad Volume: 238 bbl (5gpt acetic acid, 4 gpt surfactant, 10 gpt ammonium persulfate)
- Main Frac Volume: 2,505 bbl crosslinked gel (1,000 gal 45 lb/Mgal & 1,505 gal 40 lb/Mgal)
- Proppant: 20/40 & 16/30 mesh size Proppant tailed in with Sinterlock proppant, staged from 1 to 7 ppg
- Rate: 30 bpm

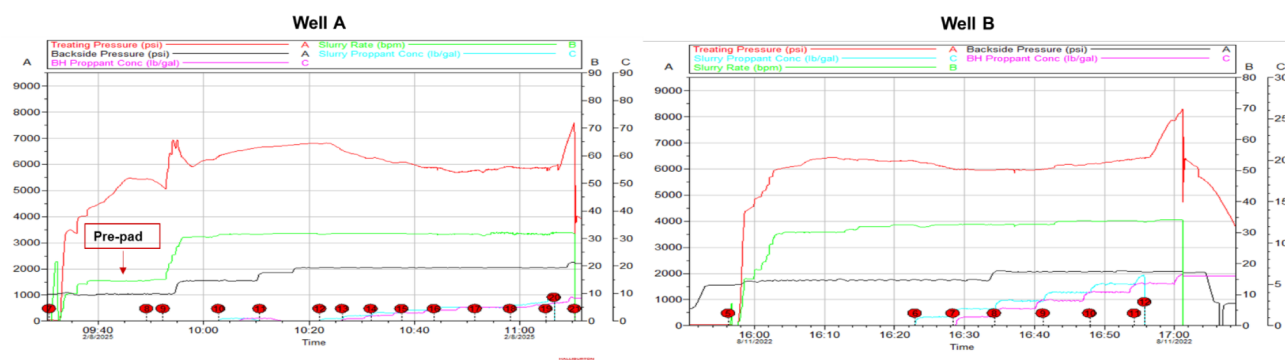


Figure 3—Job Execution

Frac Treatment Design – Well B

Well B was treated without Prepad only Xlinked fluid. Key treatment parameters included as shown in figure 3 :

- Pre-Pad Volume: NO Pre-Pad
- Main Frac Volume: 1,860 bbl crosslinked gel (1,000 gal 45 lb/Mgal & 860 gal 40 lb/Mgal)
- Proppant: 16/30 mesh size Proppant tailed in with Sinterlock proppant, staged from 1 to 6 ppg
- Rate: 30 bpm

Treatment Execution

Figure 3 shows pumping charts represent the hydraulic fracturing treatments for two offset wells, allowing for a comparative assessment of the pressure response, slurry rate, and proppant concentration throughout each operation. In the first well the treatment began with a clearly marked pre-pad stage. Treating pressure rose rapidly to a peak of approximately 7,500 psi before stabilizing for the duration of the job. The slurry rate was maintained around 30 bpm, indicating effective control of the pumping schedule, while backside pressure remained relatively steady at 3,000 psi. Proppant concentration was introduced in incremental stages, reaching a maximum of about 7 lb/gal.

In comparison, the second well showed a slightly higher peak treating pressure of around 7,800 psi. The slurry rate was marginally higher, averaging between 30 and 35 bpm, but exhibited a consistent trend throughout the treatment. Proppant concentration increased gradually as well, eventually reaching levels

similar to those in the first well. The overall pressure behavior was not as smooth as well A, which may indicate more favorable reservoir conditions or improved fracture containment. Unlike the first job, no specific pre-pad annotation was included in the chart.

The execution for both wells were smooth with no screenouts, friction lockups, or crosslink instability. Overall, both treatments were conducted with good operational control, though the similarity in pressure trends and rate behavior suggest potential similarity in formation properties, fluid design, or near-wellbore conditions. The smoother pressure response observed in well A could point to less formation damage.

No abnormal treating pressure responses were observed during the pre-pad phase, indicating full compatibility with formation and pad gel. In fact, the pressure signature during the main frac in Well A was smoother and more consistent than in prior treatments, likely due to improved near-wellbore permeability.

Post-Frac Cleanup and Flowback

The most immediate and measurable impact of the pre-pad treatment was in fluid recovery reflected in well production. This aligns with past observations where pre-conditioning treatments were linked to faster cleanup in tight formations (Fan et al., 2018). High fluid recovery not only indicates more efficient gel breaking but also reflects better displacement of frac fluid from the fracture and matrix. This is critical in the Abu Roash (G) reservoir, where low permeability and depleted pressure hinder spontaneous flowback.

The production profiles of the two offset wells offer clear insights into the effectiveness of the fracture treatment, particularly in terms of fluid cleanup and pre-pad recovery. Well A, which exhibited a more stable and elevated production rate post-fracturing, indicates a more efficient cleanup of the injected fluids, including the pre-pad stage. Shortly after the onset of production, Well A reached oil rates near 1,000 BOPD and maintained this level with only minor fluctuations. This suggests that the fracture network was effectively established and rapidly connected to the formation, allowing for early and sustained hydrocarbon flow. The strong initial response implies minimal damage or obstruction from residual gel or fluid, pointing to successful fluid recovery and clean fracture faces.

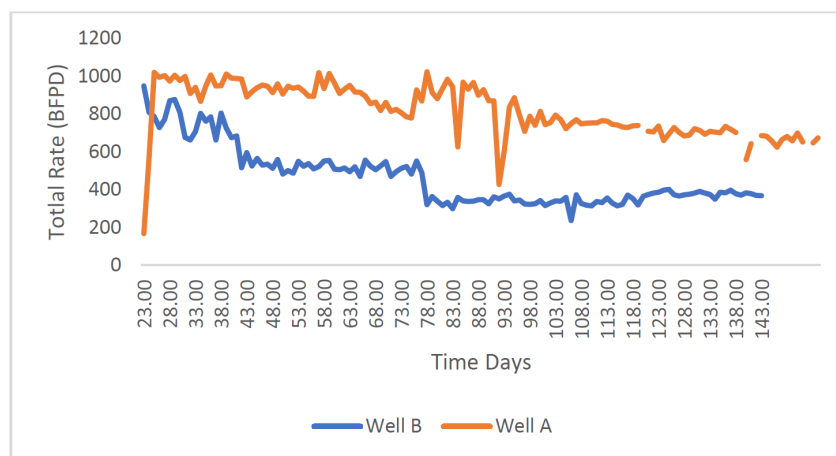


Figure 4—Production Comparison

In contrast, Well B demonstrated an early production rate similar to Well A but experienced a sharp and sustained decline within the first few days. The oil rate dropped to nearly half of its initial value and remained in the range of 450–500 BFPD throughout the remainder of the monitoring period. Both wells exhibited minimal water production, with water cuts below 2%. This trend is indicative of less effective cleanup, where residual fracturing fluids—particularly from the pre-pad stage—may not have been adequately recovered. The incomplete displacement of the fluid system likely contributed to reduced permeability and capillary trapping effects near the wellbore, inhibiting flow and resulting in premature pressure depletion.

These contrasting production behaviors underscore the critical role of effective pre-pad fluid recovery and cleanup mechanisms in enhancing early-time production. While both wells were treated with similar pumping schedules, the superior cleanup performance observed in Well A translated directly into a more favorable production outcome. This supports the hypothesis that post-fracture well performance is not solely dependent on proppant placement, but also heavily influenced by the efficiency of fluid recovery, particularly from the initial pre-pad and pad stages.

From a logistics standpoint, the pre-pad stage added approximately 20 minutes to the total pump time and increased material cost by ~5% of the total job. This is relatively minor given the uplift in production and shortened cleanup time, and compares favorably with more expensive alternatives like diverting agents, tracers, or complex fluid systems.

Results and Discussion

The results of the field trial and comparative analysis between the pre-pad-treated well (Well A) and the conventional offset (Well B) provide clear evidence of the operational and production benefits of the pre-pad breaker approach in depleted sandstone reservoirs.

Broader Implications and Literature Context

The success of this case study supports the broader applicability of pre-pad strategies in tight, low-pressure reservoirs where conventional fluid systems underperform. Similar approaches have shown promise in unconventional and tight formations worldwide, though most are applied as post-frac treatments rather than proactive pre-pads (Al Shueili et al., 2023).

Importantly, the simplicity of the chemical pre-pad system allows it to be integrated into existing operations with minimal procedural change. This contrasts with more invasive methods like coiled tubing cleanouts or remedial acid jobs that carry higher operational risk.

These results also reinforce prior findings by Palanivel et al. (2022), who emphasized the importance of sequencing and fluid compatibility in maximizing fracture conductivity and cleanup speed. The success of the Abu Roash (G) pilot underscores that performance gains can be achieved not only through advanced proppant or diversion technologies but also through smart, targeted fluid chemistry.

Conclusions and Recommendations

The implementation of a chemically aggressive pre-pad breaker system in the Abu Roash (G) sandstone reservoir has demonstrated measurable improvements in hydraulic fracturing performance under depleted conditions. Based on the field results, lab validation, and comparative analysis, several conclusions and forward-looking recommendations can be drawn.

Enhanced Fluid Recovery

Wells treated with the pre-pad approach recovered 60–70% of frac fluids within the first 72 hours, compared to <50% in offsets. This rapid cleanup shortened the flowback window, accelerated time to peak production, and reduced operational burden.

- 1. Improved Gel Breaking and Emulsion Control** The combination of a high-concentration oxidizing breaker, acetic acid, and microemulsion surfactant successfully weakened polymer structure and destabilized emulsions. This was evident in both lab filter tests and flowback fluid clarity, supporting improved regained permeability and lower near-wellbore skin.
- 2. Higher Initial Productivity**

The pre-pad-treated well exhibited a 29% increase in early oil rate compared to an offset treated with conventional fluids. These results point to cleaner fracture faces and better fluid conductivity near the wellbore.

3. **Operational Simplicity and Low Cost Impact** The pre-pad stage required minimal changes to field execution, adding roughly 30 minutes of pumping time and increasing fluid cost by just 5–7%. No additional equipment or personnel were required, making this an easy-to-adopt enhancement with immediate returns.
4. **Applicability to Other Depleted Sandstone Reservoirs** The method is not limited to the Abu Roash (G). It offers a viable solution for other tight, pressure-depleted formations where conventional clean-up strategies underperform. The system's chemical flexibility makes it compatible with most crosslinked gel fluids used in the region and globally.

The pre-pad breaker system demonstrated in the Abu Roash (G) reservoir is not limited to this specific formation. It offers a broadly applicable solution for other tight, pressure-depleted sandstone reservoirs where traditional clean-up strategies often fail due to polymer residue, water block, and emulsion challenges. The system's chemical composition featuring oxidizers, organic acid, and surfactants can be tuned for compatibility with a wide range of crosslinked gel systems used in the Middle East, North Africa, and beyond.

One potential enhancement to future field trials and diagnostics is the integration of pressure buildup tests post-fracturing. Buildup tests, conducted after a shut-in period following flowback, offer a robust method for quantifying well performance and near-wellbore skin. When a pre-pad system is used, a properly conducted buildup test can reveal whether the treatment achieved its intended purpose of reducing damage, enhancing cleanup, and improving connectivity between the fracture and the wellbore.

By analyzing the pressure response during shut-in, the skin factor can be calculated and compared it against historical or offset data. A consistently lower skin value in pre-pad-treated wells would reinforce the system's effectiveness and provide quantitative justification for its broader deployment. Additionally, buildup tests can help assess reservoir pressure restoration, which is important in depleted zones where frac efficiency is often compromised by poor pressure support.

For fields with multiple candidate intervals or phased development strategies, incorporating systematic buildup testing alongside pre-pad treatments offers a dual benefit: improving immediate well performance while building a data set that supports long-term stimulation planning. This diagnostic approach can also be used to fine-tune chemical concentrations or soak times for different reservoir conditions.

In summary, the pre-pad system is not only operationally simple and cost-effective but can also be validated and optimized further using field-acquired pressure data. Buildup analysis provides an underutilized yet powerful tool to track the impact of stimulation design choices especially in tight, damaged, or depleted formations.

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